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JC4004 – Computational Intelligence

Sessions 39 & 40: Generative AI

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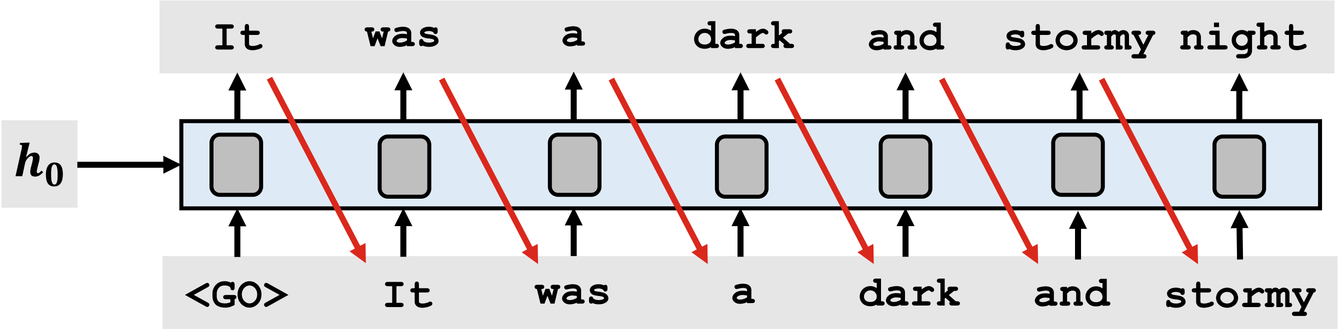
What is generative AI?

- *Generative AI* is an area of AI using *generative models* to produce text, images, speech, video, or other types of data
  - Early examples of generative AI include e.g., chatbots
  - Often (but not always), generative AI produces output in response to a textual prompt that specifies the content to be generated
- The latest generative AI models produce text, image, and speech that is difficult to distinguish from human produced content
  - Many applications, from healthcare to marketing and entertainment
  - Ethical concerns: generative AI can produce *deepfakes* for fooling people

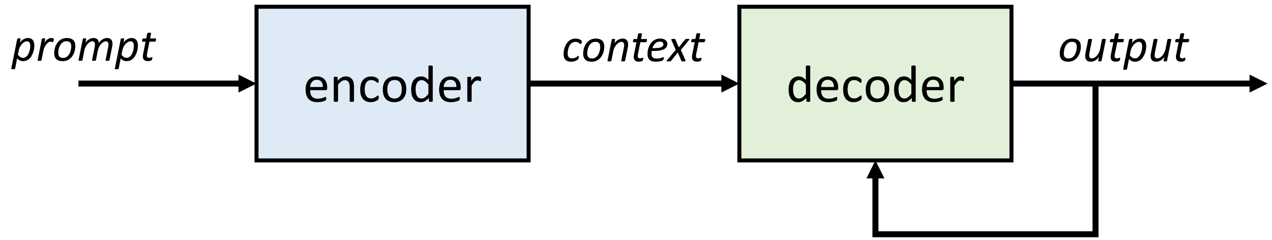
Text generation

- The most prominent example of generative AI is text generation
  - First successful text generation models were implemented using RNNs, but transformer-based models are currently the most popular models
  - S4 models (especially Mamba) are also gaining popularity
  - Mostly *autoregressive* models (i.e., output shifted and used as input)



Encoder-decoder architecture

- In principle, generative models can produce content without instructions (e.g., initialised with noise)
- However, a common practice is to use an *encoder* to set the context and *decoder* for actual content generation
  - Some examples discussed in the earlier lectures of this course



Speech synthesis

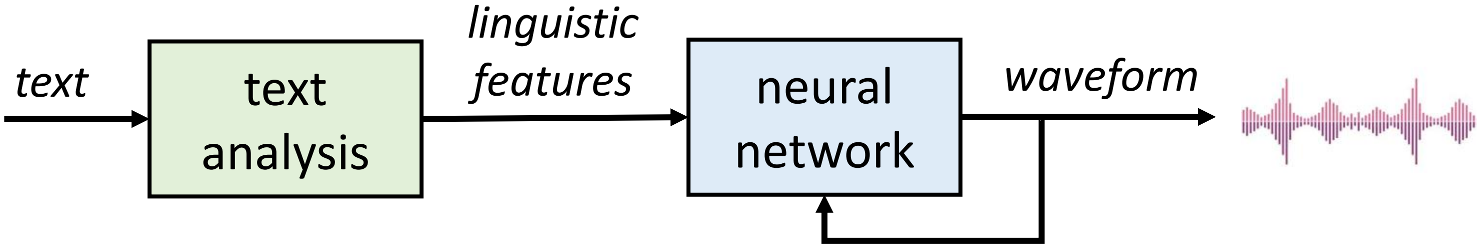
- Given input text (or some other linguistic units) Y, the goal of speech synthesis is to generate target speech X using model parameters θ

$$X = \arg\max P(X|Y, \theta)$$

- Apart from text generation, speech synthesis is also one of the most important examples of generative AI
  - Before the deep learning era, speech synthesis systems based on e.g., HMMs or Gaussian mixture models often produced “robotic” voice
  - Modern speech synthesis systems are based on artificial neural networks, and they can e.g., credibly mimic the voice of a specific person

Neural speech synthesis

- Deep neural speech synthesis models typically use a text analysis module to extract linguistic features from the input text
  - Linguistic features include information about pronunciation and prosody of individual characters to guide the speech synthesis process
  - Additional information can be also used as input to adjust voice characteristics (e.g., male/female voice etc.)



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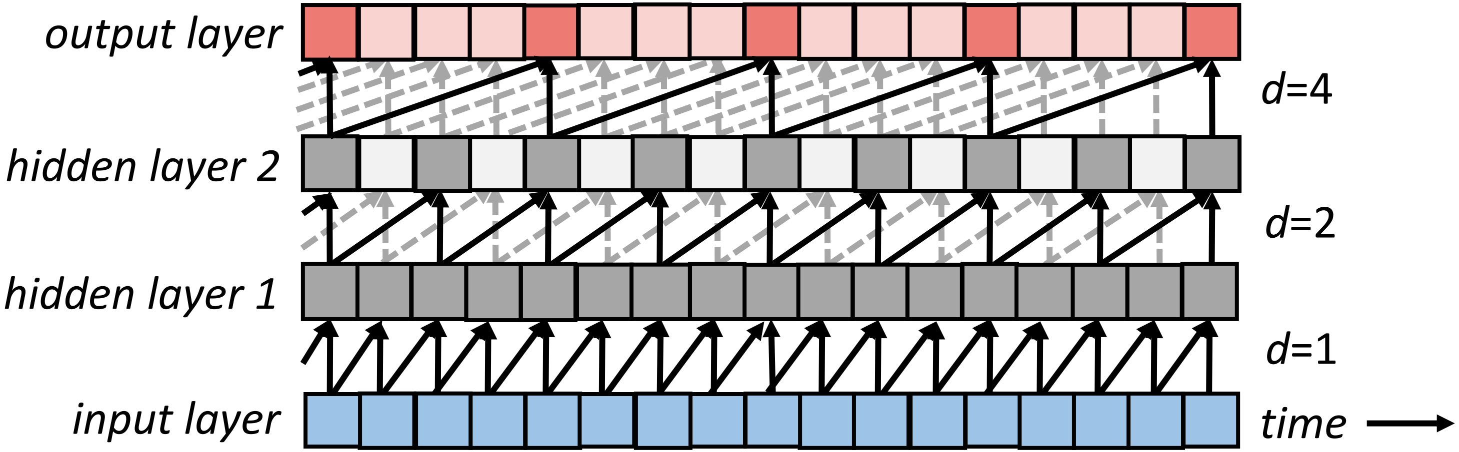
Dilated causal convolution

- Early neural speech synthesis models like *WaveNet* (2016) use dilated 1D causal convolution with dilation factor doubling from layer to layer



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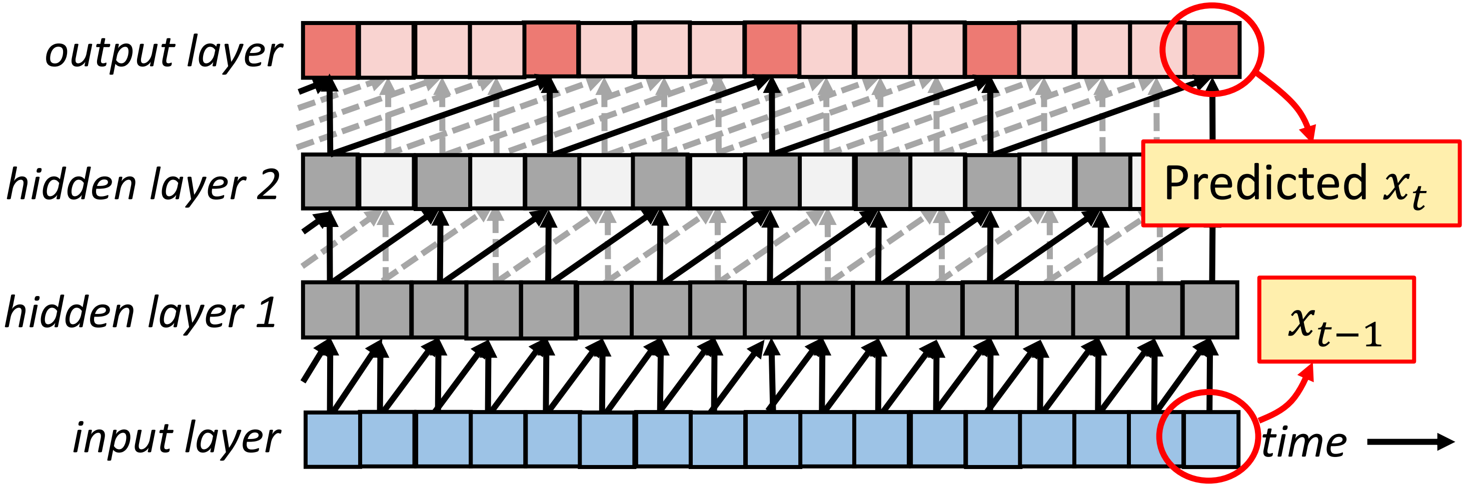
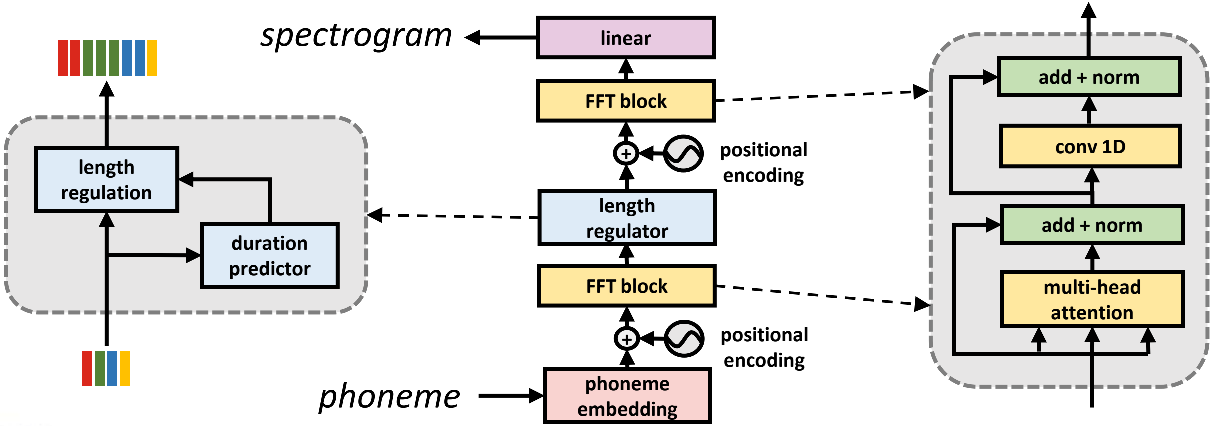


FastSpeech

- Among more recent models such as *FastSpeech*, transformers are used for generating either waveform for spectrogram
  - General architecture is somewhat similar with transformer models for NLP tasks, with some differences such as length regulation



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  - General arc
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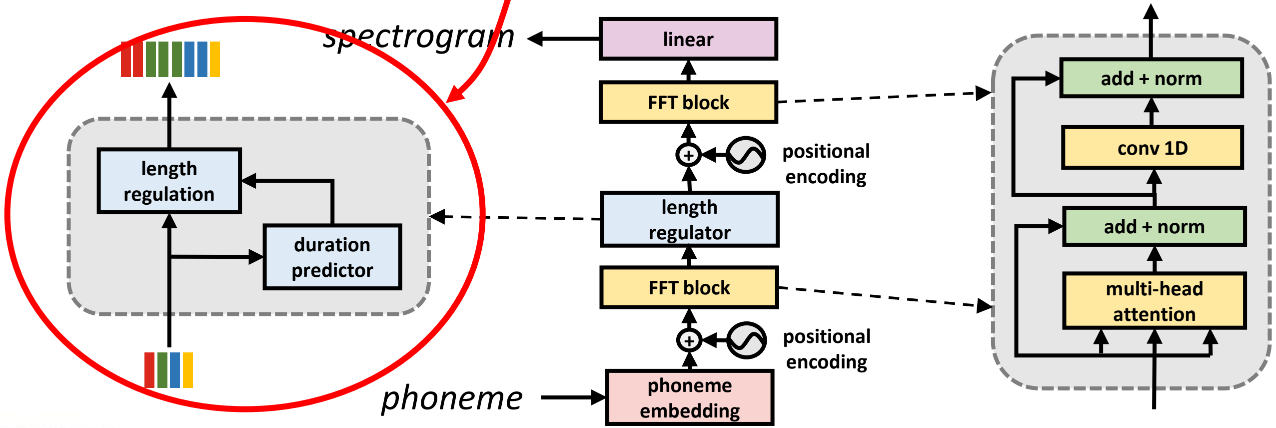
Length regulation adjusts the differences between input sequence length (phonemes) and output sequence length (waveform)



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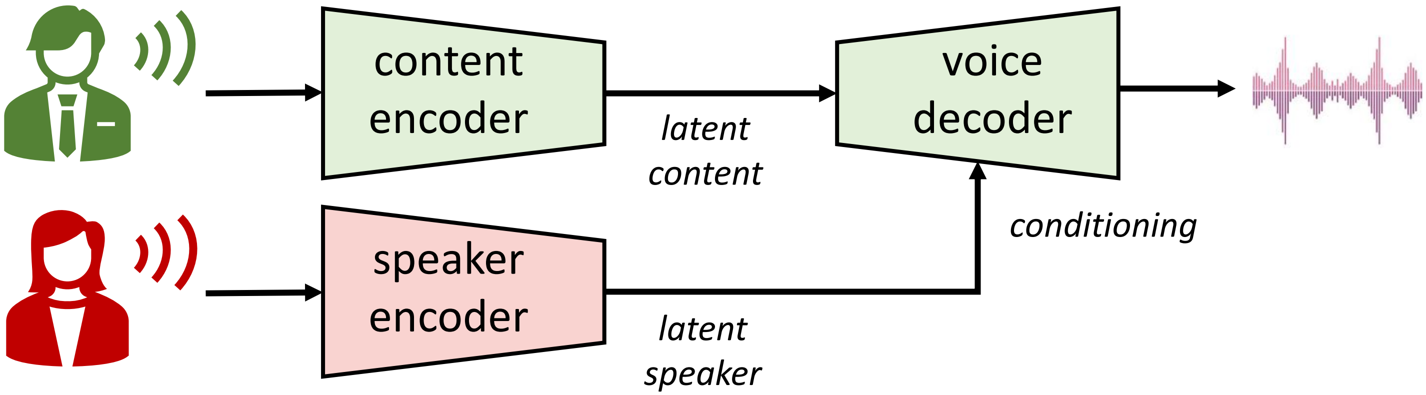
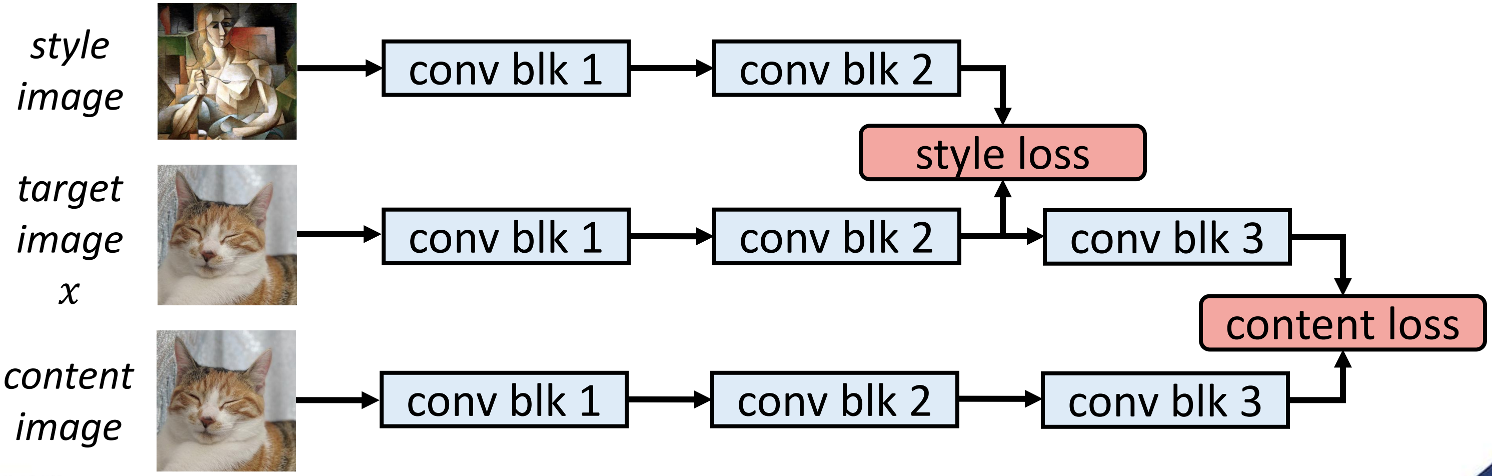
Voice conversion

- *Voice conversion* alters the style of the speech signal (e.g., voice of a specific person), but preserves the linguistic content
  - Recent techniques use content encoder to guide the content, and speaker encoder to guide the style of the generated speech



Style transfer for images

- To implement *style transfer* for images, gradient descent can be applied to the target image to fuse style and content
  - Intermediate layers represent style, and deep layers represent content



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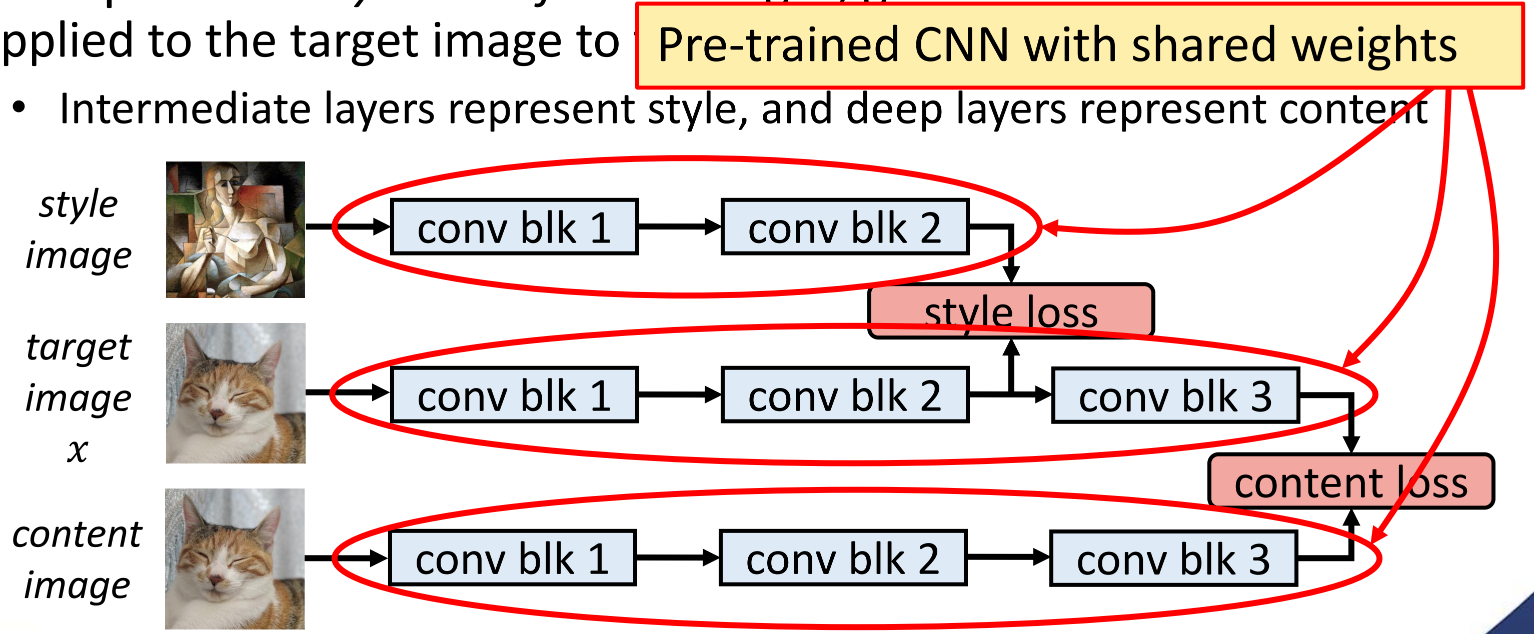
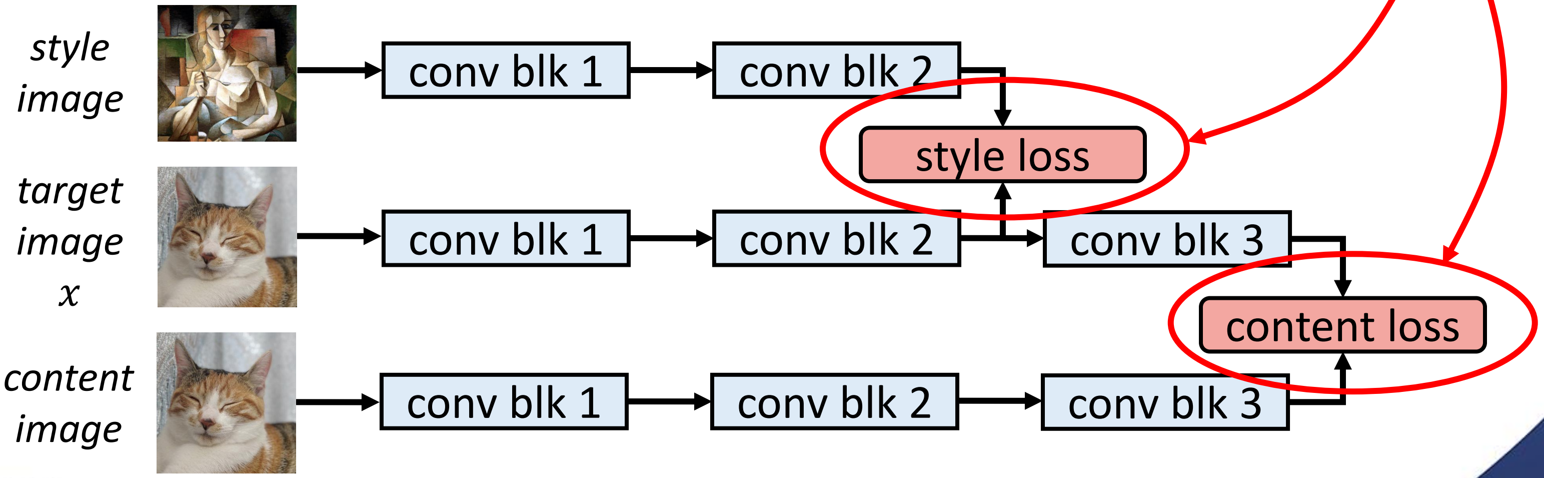


Style transfer for images

- To implement *style t*
  applied to the target i
  - Intermediate layers represent style, and deep layers represent content

Total loss is combination of style loss and content loss, for example: $L = \alpha L_{content} + \beta L_{style}$



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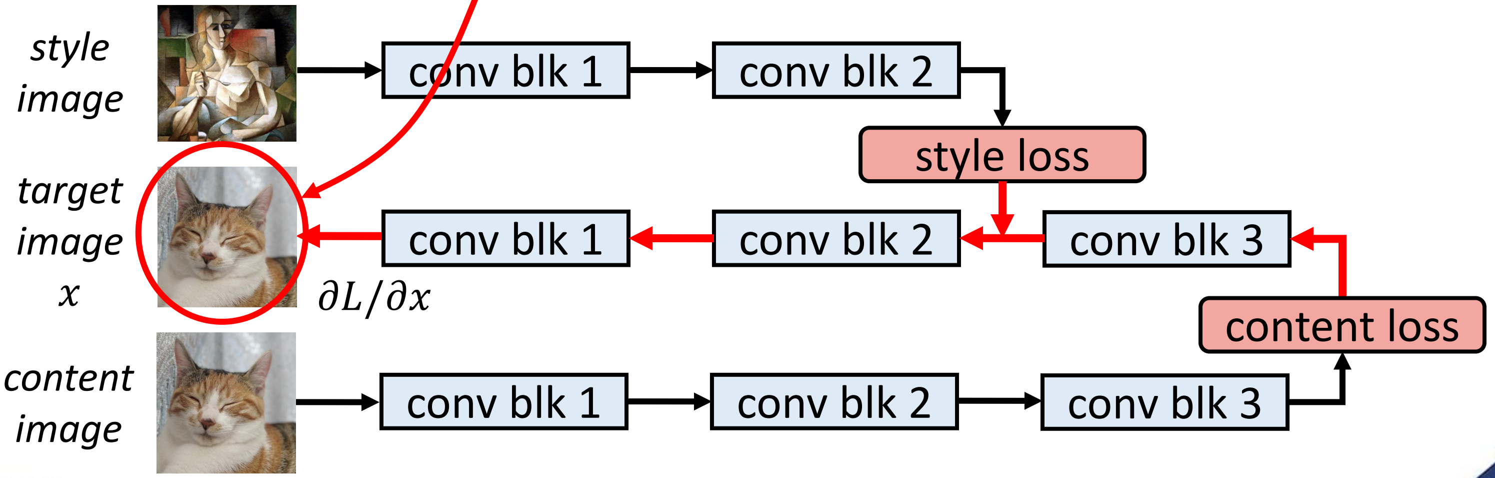
Style transfer for images

- Initial target image is the content image. Gradients of target image x in respect with total loss, i.e., $\partial L/\partial x$, are computed through backpropagation, and target image is updated by gradient descent.



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Loss functions for style transfer

- Different loss functions can be used for style transfer
  - The original paper proposed MSE for content loss and MSE between Gram matrices for style loss

$$L_{style} = \frac{1}{N}\sum_{i,j}\left(G_{style}[i,j] - G_{target}[i,j]\right)^2; \quad G = XX^T, X \in \mathbb{R}^{C\times HW}$$

  - Often, output of the last layer represents content, whereas several layers combined represent the style
  - Losses for multiple layers can be combined through weighted averaging

Style transfer example

- The update process is repeated for several iterations, and the target image eventually fuses style and content features

style image

content image

10 iterations

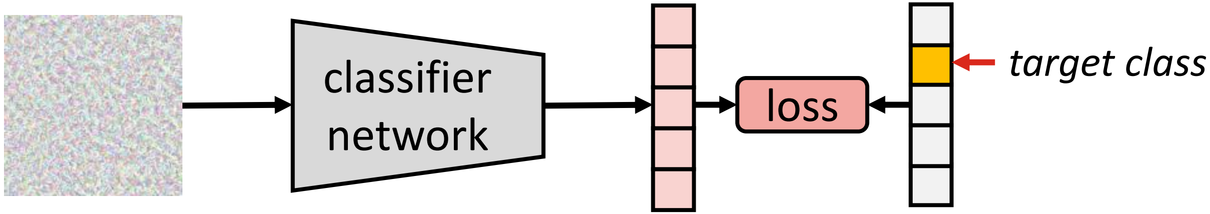
25 iterations

100 iterations

Break: any questions?

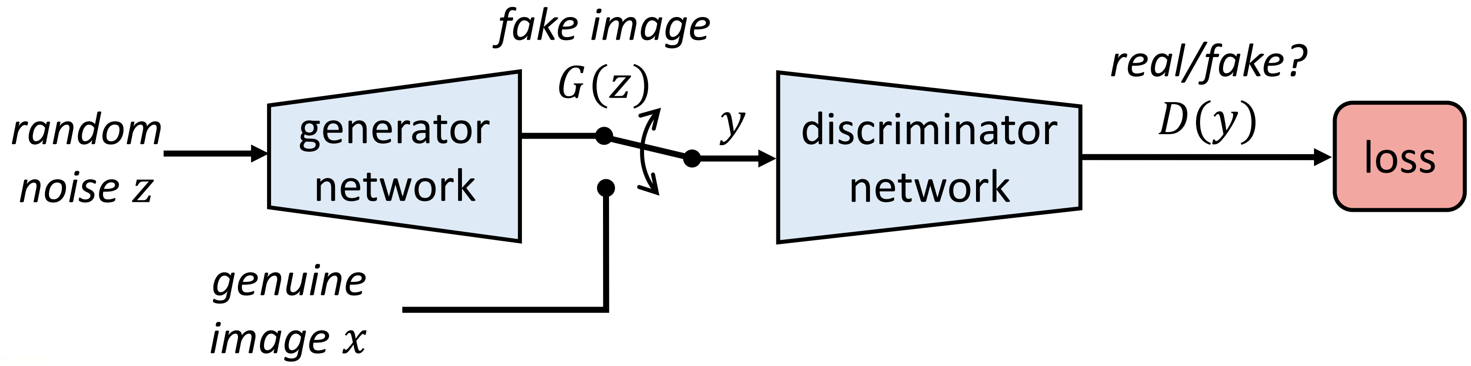
Adversarial attacks

- Inspired by style transfer, we may think that it was possible to use a pre-trained classification network to synthesise images
  - Target image initialised with noise and updated via gradient descent
  - Loss function is cross-entropy between predicted and target class
- However, this technique *does not work:* the result is a noise image, wrongly classified in the target class!
  - The method is used for *adversarial attacks* to fool classifier models



Generative adversarial networks (GAN)

- The first successful image synthesis methods were based on *generative adversarial networks* (GAN)
  - GANs consist of two networks: *generator* and *discriminator*
  - Generator network generates an image, and discriminator network aims to discriminate synthesised (fake) and genuine images



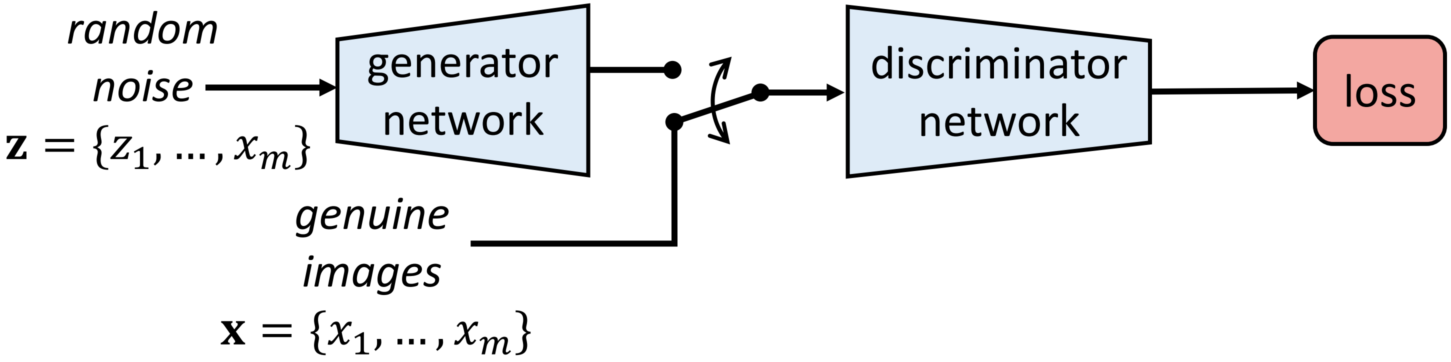
GAN loss function

- Discriminator D output is a predicted probability that the image is genuine: $D(y) \in [0,1]$
  - The goal of discriminator is to maximise the likelihood that genuine image x is predicted as real: $\max \mathbb{E}(\log(D(x))$
  - On the other hand, the goal of generator is to minimise the likelihood that fake image $G(z)$ is predicted as fake: $\min \mathbb{E}\left(\log\left(1 - D(G(z))\right)\right)$
  - The combined objective can be defined as:

$$\min_G \max_D \underbrace{\mathbb{E}\left(\log\left(D(x)\right)\right)}_{\textit{discriminator loss}} + \underbrace{\mathbb{E}\left(\log\left(1 - D(G(z))\right)\right)}_{\textit{generator loss}}$$

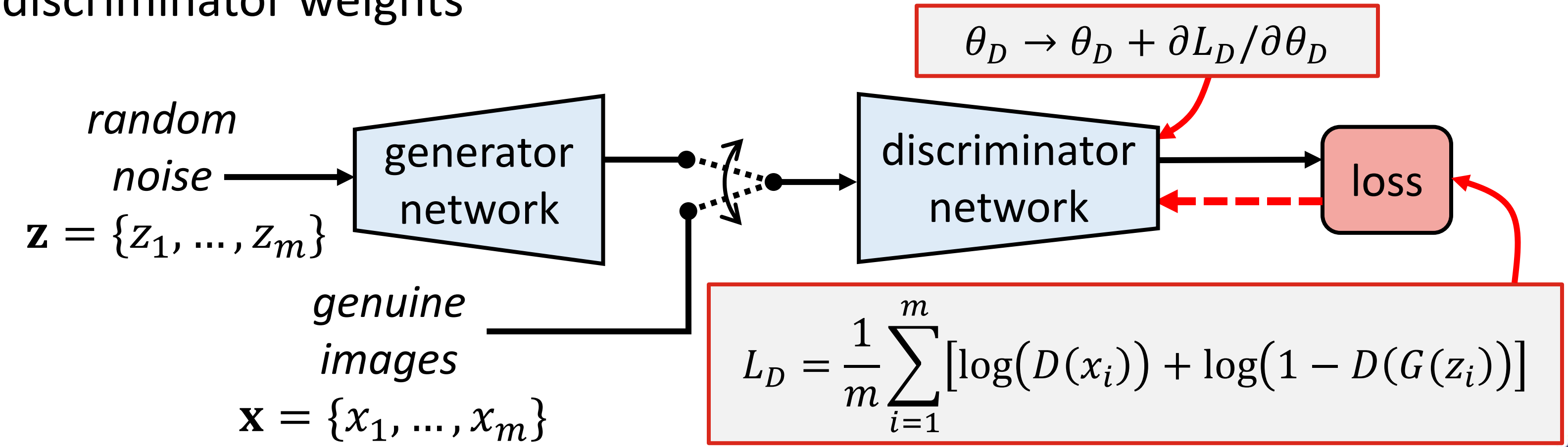
Training GAN

- When training the model, m samples of genuine images $\mathbf{x} = \{x_1, \dots, x_m\}$ and m samples of noise $\mathbf{z} = \{z_1, \dots, x_m\}$ are used



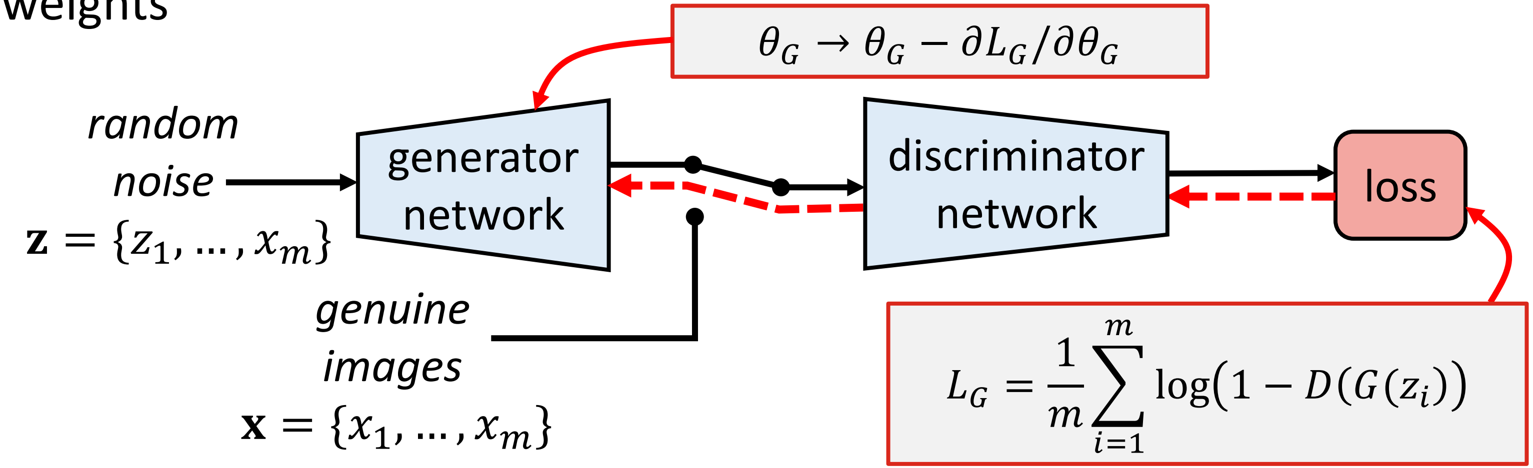
Training GAN: discriminator

- Genuine and fake samples are used to compute the combined loss
- Backpropagation and gradient descent are then used to update the discriminator weights



Training GAN: generator

- Only fake samples are used to compute the loss for generator
- Backpropagation and gradient descent used to update the generator weights

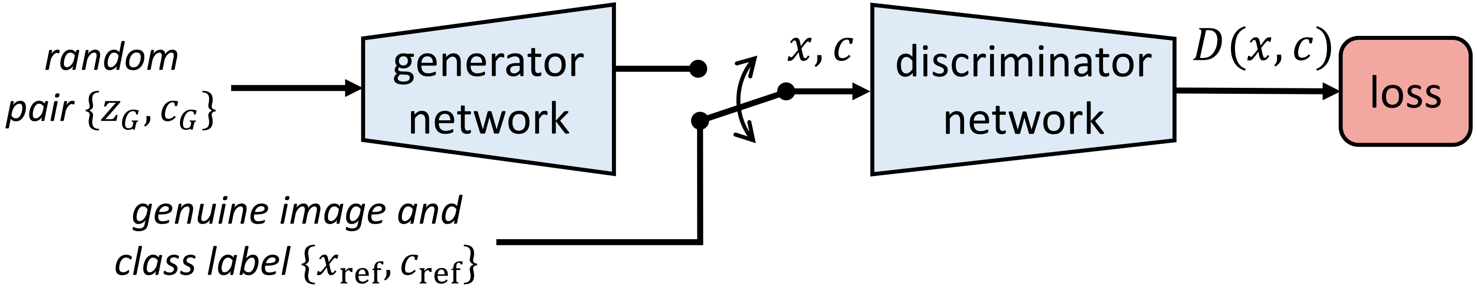


Discriminator and generator architectures

- Discriminator architectures can be based on any classification network, most typically CNN
  - Note that $\log(0) = -\infty$, so we should not allow output to be 0; sigmoid activation is often used for output, since $\mathrm{sigmoid}(x) \to 0$ as $x \to -\infty$
- Generator architectures are mostly based on transposed convolution
  - Low-dimensional noise vector or matrix upscaled into full image
- Other types of architectures such as transformers could also be used
  - A lot of different GAN models have been proposed in the literature

Conditional GAN

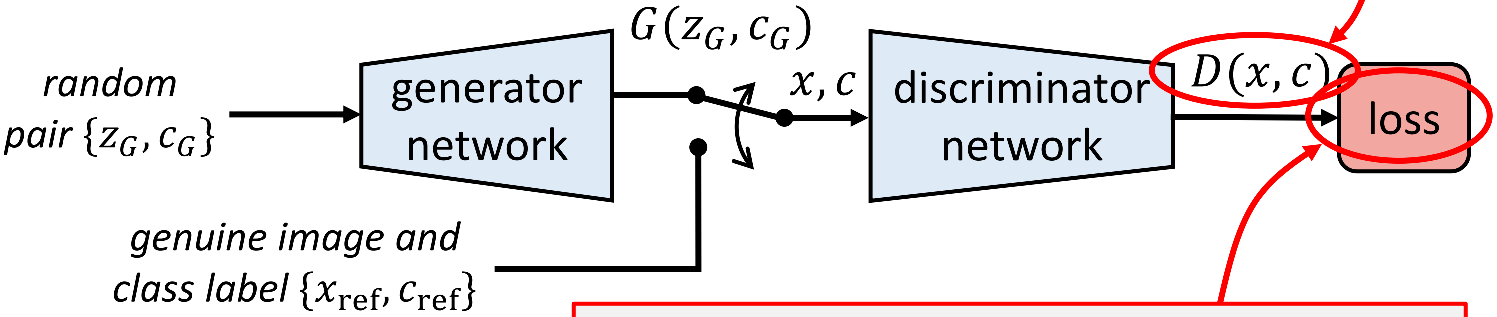
- For practical applications, it would be useful to give some instructions for image generation (e.g. text prompt)
  - A straightforward way to implement conditional GAN is to use discriminator that combines true/fake prediction with class prediction



Conditional GAN

- For practical applications, it would be useful to give some instructions for image generation (e.g. text prompt)
  - A straightforward way to implem discriminator that combines true

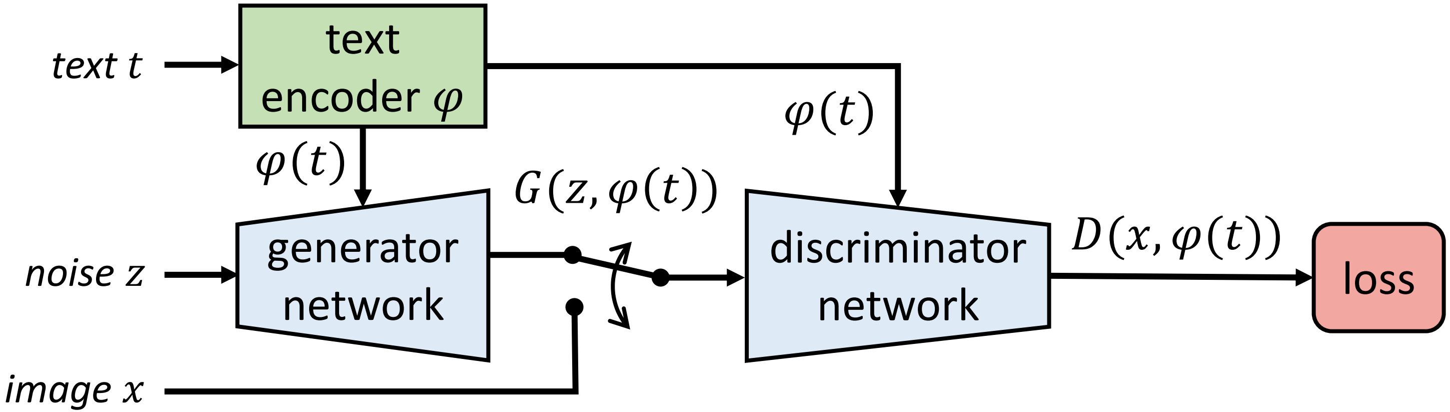
$D(x, c)$ gives probability that x is genuine, given that it belongs to class c



$$\mathbb{E}(\log(D(x_{\text{ref}}, c_{\text{ref}})) + \mathbb{E}(\log(1 - D(G(z_G), c_G))$$

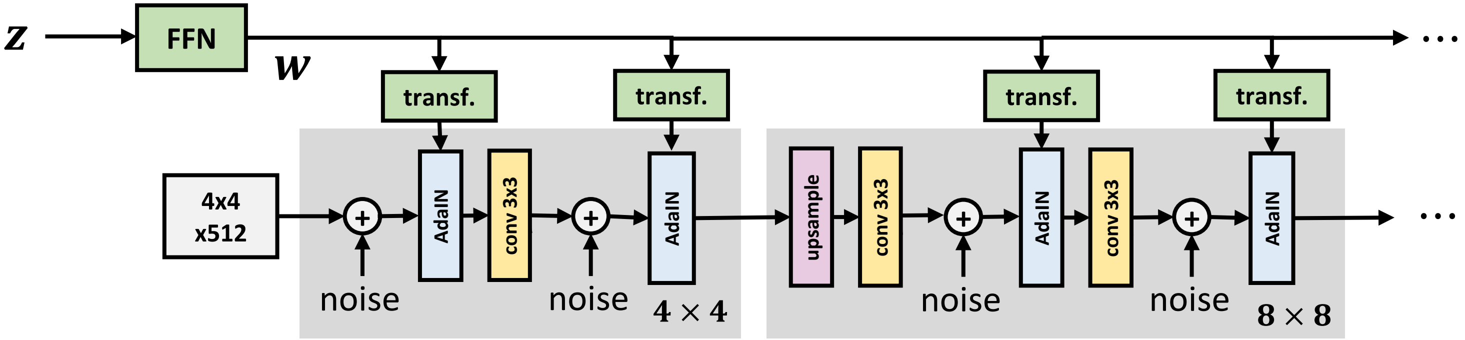
Text conditioned GAN

- Image category is not always sufficient for proper image generation
  - *Text conditioned GAN* uses a text encoder to allow more detailed instructions for image generation



StyleGAN

- StyleGAN is a family of GAN architectures developed by NVIDIA
  - Style vector $\boldsymbol{w}$, derived from latent vector $\boldsymbol{z}$, controls style characteristics such as colour and shape of details of the generated image
  - By adjusting $\boldsymbol{z}$, it is possible to variate the style of the generated images



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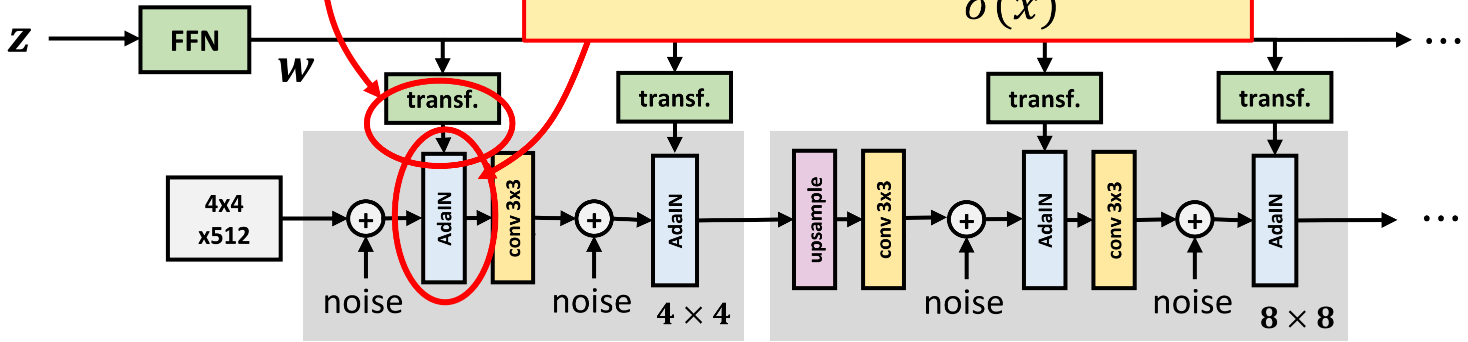
Style vector $\boldsymbol{w}$ transformed to style parameters y_s and y_b

latent vector $\boldsymbol{z}$, controls style characteristics

details of the generated image

  - By adjusting $\boldsymbol{z}$, it is pos … ated images

$$\mathrm{AdaIN}(x, y) = y_s \frac{x - \mu(x)}{\sigma(x)} + y_b$$



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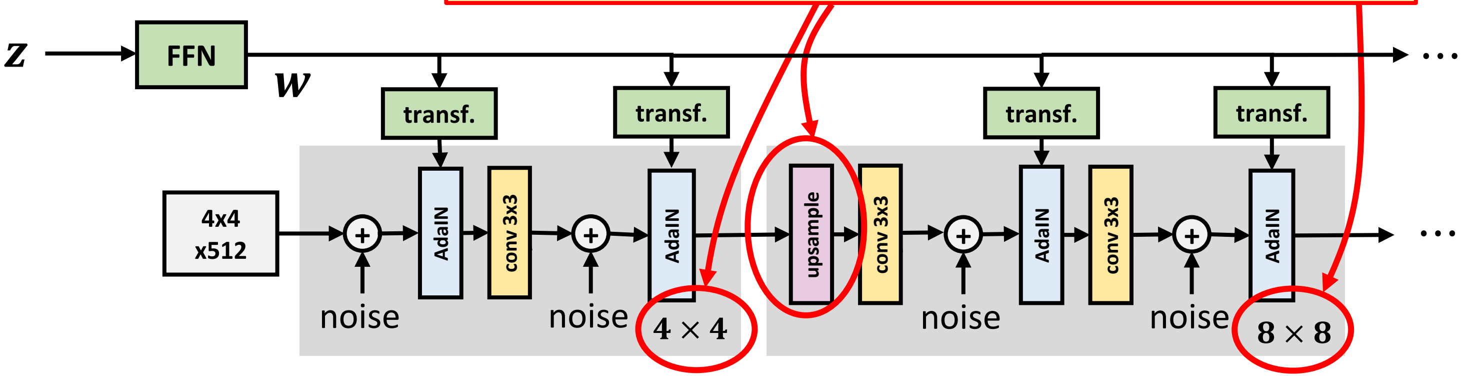
StyleGAN generates images in a progressive manner by increasing resolution block by block



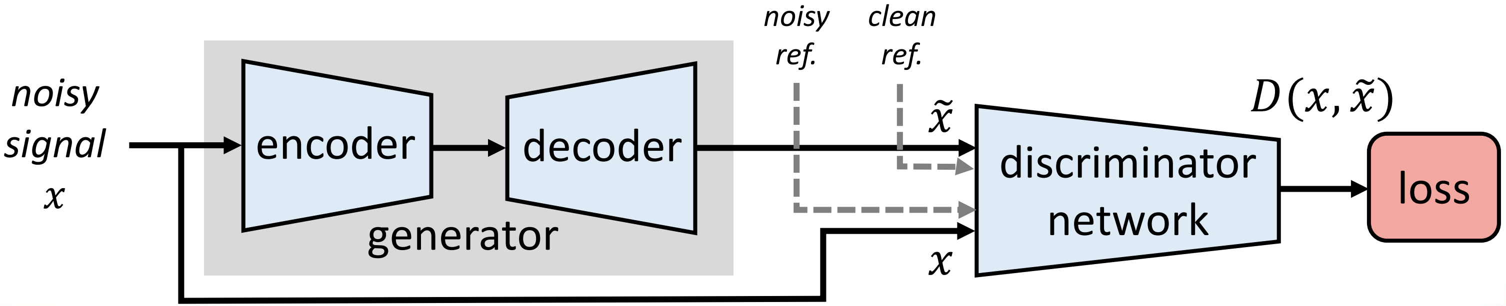
StyleGAN examples

- StyleGAN has achieved especially impressive results in generating fake portraits of human faces that look very realistic
  - Style vector can adjust parameters such as gender and age; it is also possible to mix styles
  - Added noise variates details like placement of hair, skin pores, etc.

Source: Karras et al, *"A Style-Based Generator Architecture for Generative Adversarial Networks,"* CVPR 2019.

GAN for speech and audio

- GAN can be used also for speed and audio generation
  - The basic method is similar with image GAN, but the architectures for the generators and discriminators need to be adjusted for audio data
- A relevant application of GAN is audio denoising
  - Discriminator has two inputs, noisy signal x and clean signal $\tilde{x}$, and its task is to distinguish them from each other



Summary

- *Generative AI* is a prominent area of AI, with focus on generating new content such as text, speech, or images
  - Text generation often based on autoregressive transformer decoders
  - Speech and audio generation often based on dilated 1D causal convolutions, or more recently, transformers
- *Generative adversarial networks (GAN)* is one of the most common techniques to train generative models for images and audio
  - Baseline GAN contains two jointly trained networks: *generator* to generate content, and *discriminator* to distinguish real and fake content
  - More advanced conditioned versions of GAN can be used to generate content from text prompts

Questions and discussion

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